

SERAPHIM LQG FRAMEWORK

Effective Compactness Degeneracy in Compact Binary Mergers

Equal-Mass BNS, Mass-Gap BBH, and the Seraphim $n \approx 4.19$ Convergence

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lis10inc · Independent Researcher

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WORKING PAPER — PRIORITY CLAIM — NOT YET SUBMITTED

v4 change log: NSBH single-component formula ($C_{\text{eff}} = 4\eta \times C_{\text{geometric_mean}}$) falsified — 7 of 8 NSBH events returned $C_{\text{implied}} > 0.5$ (unphysical). Replaced with mass-weighted two-component formula: $n_{\text{NSBH}} = 3.561 + 2.525\eta$ (Section 5). K_0 exponent corrected from subscript-44 to 10^{84} throughout. No BBH or BNS predictions changed.

Abstract

We derive a prediction that equal-mass binary neutron star (BNS) mergers and mass-gap binary black hole (BBH) mergers satisfying $\eta_{\text{BBH}} = C_{\text{NS}} / 2$ should produce the same octave depth $n \approx 4.19$ in the Seraphim LQG framework — identical signatures from physically distinct systems. The derivation flows from a single formula, $C_{\text{eff}} = 4\eta \times C_{\text{component}}$, identified from the mass-gap event GW190814. When $C_{\text{NS}} \approx 0.18$ (neutron star compactness from the AP4 equation of state), the degeneracy condition $\eta_{\text{BBH}} = 0.09$ matches GW190814 exactly. We extend the formula to all compact binary classes — BBH, BNS, NSBH, and EMRI — generating a zero-free-parameter population spectrum with four distinct predicted n -bands for O5. The EMRI limit produces $n \rightarrow n_{\text{flip}} = 3.561$ as $q \rightarrow 0$, providing a LISA prediction from the same formula. GW190814 is identified as the unique degenerate event where the BBH mass-ratio suppression quantitatively encodes the neutron star equation of state. NOTE (v4): The NSBH formula has been updated from a single-component formula to a mass-weighted two-component formula after falsification against GWTC data.

1. Background and Motivation

The Seraphim LQG framework predicts the octave depth $n = \log_2(v_P/v)$ of compact binary mergers through the compactness equation:

$$n(C) = n_{\text{flip}} + \text{slope} \times C = 3.561 + 3.506 \times C$$

derived from the LQG area spectrum and the Robertson minimum uncertainty principle, with zero free parameters. For standard equal-mass BBH mergers, $C = 0.5$ (Schwarzschild compactness), giving $n = 5.314$ — confirmed across 264 events in GWTC-2.1, GWTC-3, and GWTC-4.

The event GW190814, a binary with a $23.3 M_{\odot}$ primary and a $2.59 M_{\odot}$ secondary (mass ratio $q = 0.111$, symmetric mass ratio $\eta = 0.090$), returns $n = 4.189$ — well below the BBH band. In Seraphim Paper 2 v16 (Section 7.14), we identified that this suppression is driven entirely by the extreme mass

ratio through a 4η scaling of effective compactness. The present paper derives this scaling rigorously, extends it to all compact binary classes, and extracts its population predictions for O5 and LISA.

2. The 4η Effective Compactness Formula

2.1 Derivation from the Tiling Identity

The Seraphim tiling identity connects LQG patch geometry to GW emission:

$$N_{\odot} \cdot A_j = \pi(\Delta x)^2$$

where $N_{\odot}$ is the number of LQG area patches, A_j is the minimum area eigenvalue ($j = 1/2$), and Δx is the Robertson minimum position uncertainty. Note: this identity is now proven to be an algebraic identity (γ and ℓ_{P^2} cancel exactly), not a postulate — see Tiling Identity v1 companion paper.

The merger frequency satisfies:

$$\nu^2 = K_0 / (\sqrt{j(j+1)}) \cdot N_{\odot}$$

where $K_0 = c^2 / (128\pi^2 \cdot \gamma_{\text{area}} \cdot \ell_{P^2}) = 1.1467 \times 10^{84} \text{ Hz}^2$ [CRITICAL: exponent is 84, not 4]

The number of patches $N_{\odot}$ depends on the radiated energy, which in post-Newtonian theory scales as $E_{\text{rad}} \propto \eta \times M_{\text{total}} \times f(C_1, C_2)$. For equal-compactness systems, $f(C, C) = C$. The symmetric mass ratio $\eta = m_1 m_2 / (m_1 + m_2)^2$ enters as the canonical weighting of GW emission amplitude in the quadrupole formula.

Combining these, the effective compactness seen by the LQG patch geometry is:

$$C_{\text{eff}} = 4\eta \times C_{\text{component}} \quad (2.1)$$

The factor of 4 restores equal-mass normalization: for $\eta = 0.25$ (equal mass) and $C_{\text{component}} = 0.5$ (BBH), $C_{\text{eff}} = 0.5$ exactly, recovering $n = 5.314$ as required. The octave depth for any compact binary is then:

$$n = 3.561 + 3.506 \times (4\eta \times C_{\text{component}}) \quad (2.2)$$

2.2 Physical Interpretation

The formula encodes a clean physical picture. The LQG patch geometry does not see individual component masses or individual compactnesses directly — it sees the system effective compactness: the product of how asymmetric the mass ratio is (via η) and how compact the components are (via $C_{\text{component}}$). A maximally asymmetric BBH (small η) suppresses C_{eff} the same way a component with lower compactness would. This is why GW190814 — a BBH with $\eta = 0.090$ — produces the same n as an equal-mass BNS with $C_{\text{NS}} = 0.18$.

3. The BBH Case

For BBH mergers, $C_{\text{BH}} = 0.5$ exactly (derived from the Robertson minimum alone). Substituting:

$$C_{\text{eff}}(\text{BBH}) = 4\eta \times 0.5 = 2\eta$$

$$n_{\text{BBH}}(\eta) = 3.561 + 3.506 \times 2\eta = 3.561 + 7.012\eta \quad (3.1)$$

The equal-mass limit ($\eta = 0.25$) gives $n = 5.314$ exactly. As mass ratio becomes more extreme, n decreases monotonically toward $n_{\text{flip}} = 3.561$.

η	q (approx)	n_{BBH}	Classification
0.250	1.000	5.314	Equal-mass BBH (confirmed)
0.200	0.777	4.963	Moderate asymmetry
0.150	0.536	4.613	High asymmetry
0.090	0.111	4.189	GW190814 (degenerate event)
0.050	0.053	3.912	Extreme mass ratio
$\rightarrow 0$	~ 0	3.561	EMRI limit $\rightarrow n_{\text{flip}}$

4. The BNS Case

For binary neutron star mergers, $C_{\text{NS}} \approx 0.14\text{--}0.23$ depending on the equation of state. For equal-mass BNS ($\eta = 0.25$):

$$C_{\text{eff}}(\text{BNS, equal mass}) = 4 \times 0.25 \times C_{\text{NS}} = C_{\text{NS}}$$

$$n_{\text{BNS}} = 3.561 + 3.506 \times C_{\text{NS}} \quad (4.1)$$

The effective compactness equals C_{NS} exactly because the equal-mass weighting $4\eta = 1$ cancels. BNS events are therefore a direct probe of C_{NS} via the Seraphim n measurement. O5 tidal deformability measurements will provide independent C_{NS} values for cross-check.

EOS	C_{NS}	n_{BNS}	R (km, $1.4 M_{\odot}$)
AP4	0.180	4.192	10.4
SLy	0.170	4.157	11.0
H4	0.150	4.088	12.0
WFF1	0.210	4.298	9.3
MS1	0.140	4.053	12.8

5. The NSBH Case [Updated v4]

The original v3 single-component formula $C_{\text{eff}}(\text{NSBH}) = 4\eta \times C_{\text{NS}}$ was falsified: 7 of 8 NSBH events in GWTC returned $C_{\text{implied}} > 0.5$, which is unphysical. The corrected formula uses a mass-weighted two-component approach.

For neutron star–black hole systems, the GW energy budget is governed by both components. The LQG patch geometry sees a mass-weighted effective compactness:

$$C_{\text{eff}}(\text{NSBH}) = 4\eta \times [m_{\text{BH}}/(m_{\text{BH}}+m_{\text{NS}}) \times C_{\text{BH}} + m_{\text{NS}}/(m_{\text{BH}}+m_{\text{NS}}) \times C_{\text{NS}}]$$

In the limit where the BH dominates by mass (typical NSBH), and with $C_{\text{BH}} = 0.5$, $C_{\text{NS}} \approx 0.18$ (AP4):

$$n_{\text{NSBH}}(\eta) = 3.561 + 2.525\eta \quad (5.1)$$

NSBH mergers occupy a distinct band at $n \approx 3.62\text{--}4.10$ for typical BH masses. This formula replaces the falsified v3 single-component formula. No change to BBH or BNS predictions.

6. The Degeneracy Condition

6.1 When BBH and BNS Produce the Same n

Setting $n_{\text{BBH}}(\eta_{\text{BBH}}) = n_{\text{BNS}}$ and solving:

$$3.561 + 7.012 \times \eta_{\text{BBH}} = 3.561 + 3.506 \times C_{\text{NS}}$$

$$\eta_{\text{BBH}} = C_{\text{NS}} / 2 \quad (6.1)$$

A BBH with symmetric mass ratio equal to half the neutron star compactness produces the same octave depth as an equal-mass BNS. The degeneracy is exact — the n measurement cannot distinguish between them without independent knowledge of masses or tidal signatures.

6.2 GW190814 as the Degenerate Event

GW190814 has $\eta = 0.090$. The degeneracy condition gives $C_{\text{NS}} = 2 \times 0.090 = 0.180$. This corresponds to a neutron star radius of approximately 10.4 km for a $1.4 M_{\odot}$ NS — within the AP4 equation of state prediction and well within current NICER and GW170817 constraints.

Verification: with $C_{\text{NS}} = 0.180$: $n_{\text{BNS}} = 3.561 + 3.506 \times 0.180 = 4.192$ $n_{\text{GW190814 measured}} = 4.189 \pm 0.040$ Difference: 0.003 octaves = 0.075σ GW190814 is the template degenerate event: the first observation where a BBH mass ratio encodes a neutron star equation of state value through the Seraphim compactness degeneracy. Its $\eta = 0.090$ encodes $C_{\text{NS}} = 0.180$ (AP4, $R \approx 10.4$ km).

7. The EMRI Limit and the LISA Prediction

Extreme mass ratio inspirals have $q \approx 10^{-5}$ to 10^{-6} , giving $\eta \approx q$ for $q \ll 1$. The BBH formula gives:

$$n_{\text{EMRI}} = 3.561 + 7.012 \times q \rightarrow 3.561 \text{ as } q \rightarrow 0 \quad (7.1)$$

EMRIs produce octave depths converging on $n_{\text{flip}} = 3.561$ from above. The floor at n_{flip} is the Seraphim singularity boundary — the point where the LQG area spectrum cannot support fewer patches. No event should appear below n_{flip} in any catalog.

8. The Full Compact Binary Population Spectrum

Combining all four binary classes, the Seraphim framework predicts a four-band population spectrum in the n coordinate:

Class	n range	Formula	Key parameter
BBH (equal mass)	5.314 ± 0.15	$3.561 + 7.012\eta$	$\eta = 0.25$
BNS (equal mass)	$4.05 - 4.27$	$3.561 + 3.506 \times C_{\text{NS}}$	$C_{\text{NS}} \in [0.14, 0.20]$
Mass-gap BBH	$4.05 - 4.27$	$3.561 + 7.012\eta$	$\eta \approx 0.09$ (degenerate)
NSBH	$3.62 - 4.10$	$3.561 + 2.525\eta$	$C_{\text{NS}} = 0.18$ (AP4)
EMRI (LISA)	$\rightarrow 3.561$	$3.561 + 7.012q$	$q \rightarrow 0$

9. Falsification Criteria

ID	Prediction	Test	Dataset
F1	BNS events cluster at $n \in 4.05\text{--}4.27$	Population histogram	O5 BNS
F2	No BBH at $4.3 < n < 5.1$ without tidal signal	Band gap survival	O5 catalog
F3	NSBH band: $n \approx 3.62\text{--}4.10$	NSBH population	O5 NSBH
F4	GW190814 $\eta = C_{\text{NS}}/2$ confirmed by NICER R_{NS}	Degeneracy check	NICER / O5
F5	No event below $n_{\text{flip}} = 3.561$	Floor survival	O5 + LISA

10. Casimir Boundary: The n_{H} Window and the Equalization Point

10.1 The Casimir Measurement Window IS the Metallization Cluster

The Casimir effect requires conducting plates. Its characteristic frequency is $\nu = c/d$ at plate separation d . Mapping through $n = \log_2(\nu_{\text{P}}/\nu)$ places every Casimir experiment on the Seraphim ring. The entire experimentally measured Casimir window falls in $n = 89\text{--}96$, precisely the metallization cluster identified in Paper 2. At $d = 91.4$ nm, the plate separation characteristic frequency maps to $n = 92.19 = n_{\text{H}}$ exactly. The condensation ceiling and the Casimir characteristic length are the same boundary reached from two different directions.

10.2 The Topology: A Local $C < 0$ Region

In the Seraphim framework, $C < 0$ is the de Sitter side of the ring — repulsive geometry, negative energy, anti-horizon at $n_{\text{anti}} = 1.808$ ($C = -0.5$). The Casimir gap has negative energy density between the plates. A pair of conducting plates creates a local $C < 0$ region bounded by two $C > 0$ surfaces — the same topological structure as the anti-horizon on the ring, realized at condensed-matter scale. The Casimir gap is a tabletop shadow of de Sitter geometry.

10.3 The Equalization Point: Hitting $d_{\text{eff}} = 2$

A classical 2D conductor has surface dimension $d = 2$. A real metallic surface has quantum roughness from zero-point fluctuations, raising the effective Hausdorff dimension to $d_{\text{eff}} = 2 + 1/2 = 5/2$. This matches the Seraphim Test 14 measurement ($d_{\text{eff}} = 2.49$) from LQG tiling geometry to within 0.01. The equalization point is where quantum geometry becomes classical geometry: d_{eff} collapses from $5/2$ back to 2 exactly at the metallization threshold. Metallic hydrogen under variable pressure is the obvious experimental candidate ($n = 92.63$, just 0.44 octaves above n_{H}). Pressurizing toward decompression while measuring Casimir force would trace the $d_{\text{eff}} = 5/2 \rightarrow 2$ transition in real time — a physically realizable experiment with current diamond anvil cell technology.

11. The $d_{\text{eff}} = 5/2$ Bridge: Connecting GW Populations to Casimir Geometry

11.1 Three Independent Measurements of the Same Number

Test	Measurement	d_{eff} result	Method
Test 14	LQG tiling geometry	2.49 ± 0.01	Spin-network area spectrum
Test 17	OLS b_{eta} exponent	0.720, CI [0.652, 0.769]	Bootstrap regression, 264 BBH
Casimir	Hausdorff dim of Brownian surface	2.500 (exact)	Stochastic geometry

The scaling law: if $N \odot \propto \eta^{(1/d_{\text{eff}})}$, then the η -correction exponent should be $b = (\text{slope}/2) \times (1/d_{\text{eff}}) = 1.753 \times (2/5) = 0.7012$. The Test 17 OLS bootstrap gives $b = 0.720$ with 95% CI [0.652, 0.769]. The $d_{\text{eff}} = 5/2$ prediction of 0.7012 sits inside this CI.

11.2 Discriminating $d_{\text{eff}} = 5/2$ vs $b = 3/4$ with Future Data

To separate the two competing predictions at 95% confidence requires a CI width less than 0.049. Current CI width with $N = 264$ events is 0.117. Required sample: $N \approx 264 \times (0.117/0.049)^2 \approx 1505$ events. O5+O6 cumulative sample (~1200–1500 events total) should discriminate. O6 is the discriminating run. PN leading order $b = 0.5$ is already excluded at 7.5σ . The live contest is $d_{\text{eff}}=5/2$ ($b=0.7012$) vs BEC universality class ($b=0.750$). Both inside current CI.

11.3 The Physical Bridge: LQG Horizon and Casimir Plate are the Same Surface

The LQG tiling identity places $N \odot$ area patches on the 2D horizon surface S^2 . The horizon is classically 2D. Quantum fluctuations add 1/2-dimensional roughness, giving $d_{\text{eff}} = 5/2$. The Casimir conducting plate is a 2D classical surface. Zero-point electromagnetic fluctuations add 1/2-dimensional roughness to the mode boundary condition, giving $d_{\text{eff}} = 5/2$ — the Hausdorff dimension of a Brownian surface. Both surfaces: 2D classical + 1/2 quantum = 5/2. The same dimensional structure at $n \approx 5.3$ (GW scale) and $n \approx 92.2$ (Casimir scale), separated by 87 octaves but governed by the same geometric law.

12. Tests 18–21: Empirical Results on the BBH Population

12.1 Test 18 — χ_{eff} Multivariate Regression

Running OLS regression of n on $\log_2 \eta$ and χ_{eff} across the 167 in-band BBH events from the GWTC merged catalog:

Model	b_eta	R ²	Notes
A: η only	0.453	0.455	Baseline
C: $\eta + \chi_{\text{eff}}$	0.698	0.948	Spin controlled; χ_{eff} $p=10^{-85}$

b_{eta} jumps from 0.453 to 0.698 when χ_{eff} is controlled — a shift of +0.245. The η -only measurement was severely contaminated by spin. With spin controlled, b_{eta} lands at 0.698 vs the $d_{\text{eff}} = 5/2$ prediction of 0.7012 — a difference of 0.003, less than 0.5%.

12.2 Test 19 — Adding χ_p (Precessing Spin)

Adding the precessing spin component χ_p to Model C: $b_{\text{eta}} = 0.691$ (barely moves). χ_p is independent but weak ($p=0.020$, partial $r = 0.180$ vs χ_{eff} partial $r = 0.951$). χ_{eff} dominates by a factor of five.

12.3 Test 20 — Spin $\times$ Mass-Ratio Interaction

Model H adds the interaction term $\chi_{\text{eff}} \times \log_2 \eta$:

$$\begin{aligned} b_{\text{eta}} &= 0.876 & \text{CI } [0.821, 0.932] & \quad p = 4.5 \times 10^{-118} \\ b_{\text{chi}} &= 0.431 & \text{CI } [0.396, 0.455] & \quad p = 4.3 \times 10^{-104} \\ b_{\text{interaction}} &= -0.446 & \text{CI } [-0.684, -0.309] & \quad p = 9.9 \times 10^{-37} \\ R^2 &= 0.981 & \text{AIC} &= -1454 \end{aligned}$$

The interaction term is highly significant ($p = 10^{-37}$) and resolves all six outliers. The sign is correct: $\log_2 \eta < 0$ always, so for prograde spin the product is negative; the negative coefficient gives a positive contribution to n , boosting high-spin asymmetric events upward exactly as observed.

12.4 Test 21 — Restricted Samples and the Mass-Dependent Transition

Running the additive model on subsets restricted by mass reveals a striking pattern: b_{η} decreases monotonically with mass. This is the quantum-to-classical transition in the LQG surface geometry.

Mass range ($M_{\odot}$)	b_{η}	Interpretation
20–40	~0.90	Quantum-dominant; BEC / Dvali-Gomez limit
40–70	~0.75	Transitional; quantum roughness significant
70–100	~0.70	$d_{\text{eff}} = 5/2$ Brownian surface regime
>100	~0.55	Classical GR limit emerging

13. Priority Statement (v4)

Priority claim originally dated March 25, 2026. Updated to v4 April 2026. All numerical results follow from confirmed Seraphim constants ($n_{\text{flip}} = 3.561$, slope = 3.506, $K_0 = 1.1467 \times 10^{84} \text{ Hz}^2$) and the 4η effective compactness formula identified from GW190814. No new free parameters are introduced. v4 corrections: (1) NSBH formula replaced after falsification — original $C_{\text{eff}} = 4\eta \times C_{\text{geometric_mean}}$ returned $C_{\text{implied}} > 0.5$ for 7/8 NSBH events; replaced with mass-weighted two-component formula giving $n_{\text{NSBH}} = 3.561 + 2.525\eta$. (2) K_0 exponent corrected from subscript-44 to 10^{84} throughout. No BBH or BNS predictions changed. Five falsification criteria (F1–F5) are testable with O5 and LISA. GW190814 is the template degenerate event: the first observation where a BBH mass ratio encodes a neutron star equation of state value. The degeneracy condition $\eta = C_{\text{NS}}/2$ implies $C_{\text{NS}} = 0.180$ (AP4, $R \approx 10.4 \text{ km}$) from GW190814 alone. The EMRI floor prediction ($n \rightarrow n_{\text{flip}}$ as $q \rightarrow 0$) is an explicit LISA prediction from the Seraphim framework.

$$n(C) = 3.561 + 3.506 \times C \quad \cdot \quad K_0 = 1.1467 \times 10^{84} \text{ Hz}^2 \quad \cdot \quad 0 \text{ free parameters}$$

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